

Customer Revenue Prediction Using Marketing Data

Dataset

Marketing and Product Performance Dataset

Source: Kaggle

Link: <https://www.kaggle.com/datasets/imranalishahh/marketing-and-product-performance-dataset>

Approximately 10,000 rows with multiple features.

Target Variable

Customer revenue level (High / Low)

Task Type

Classification

Problem Statement

Understanding the key factors that influence customer revenue is critical for optimizing marketing strategies and improving business performance. This project aims to build a machine learning model that can predict revenue level based on marketing and product-related features.

Research Questions

What is the performance of a baseline model?

Which model performs better?

Does feature scaling make a difference?

Which features are most important?

Does hyperparameter tuning improve model performance?

How well does the model perform overall?

What insights can be drawn from the analysis?

Methodology

Data cleaning and preprocessing

Creating target variable

Encoding categorical data

Splitting into train/test sets

Model training and comparison

Feature scaling

Hyperparameter tuning

Evaluation using metrics

A subset of the dataset was used to improve computational efficiency while maintaining representative data distribution.

Models Used

Logistic Regression

Random Forest

Evaluation Metrics

Accuracy, Precision, Recall, F1-score

Expected Outputs

Model comparison graphs

Feature importance chart

Confusion matrix

Tables in CSV format

Conclusion

The results of this study aim to support data-driven decision-making in marketing and product optimization.